SCCCI × NYP AI INNOVATION CHALLENGE

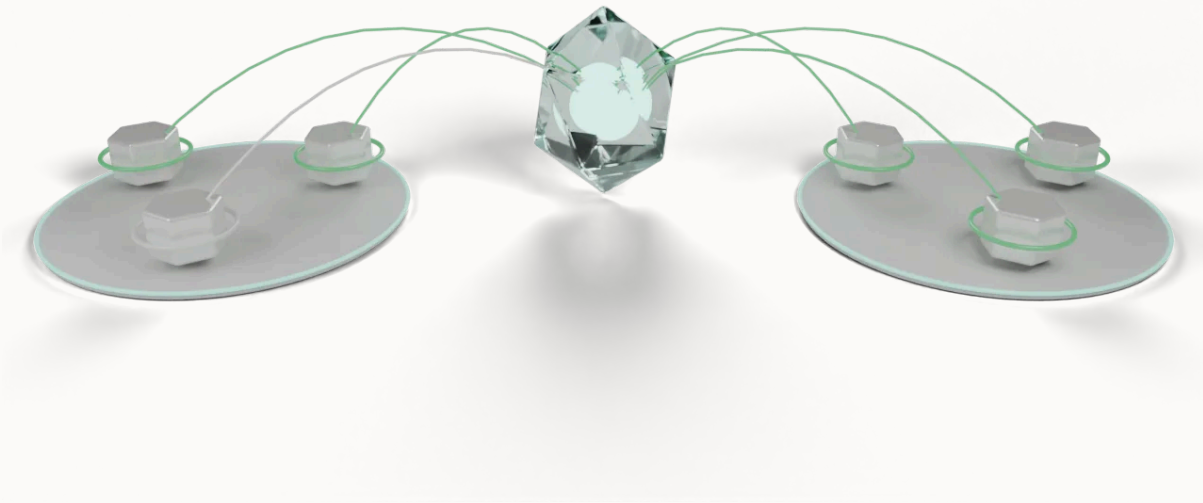
# HuiTong 汇通

An integration and workflow layer for Organisation E's HR systems

Richie · Iok Teck · Azri · Angelyn

Problem Statement 17 · IT2134 IT Innovation Project

Nanyang Polytechnic · AY2026 Semester 2



THE PROBLEM

# Mei Ling is the integration

A Singapore HR executive, on the second working day of the month

**08:45**
Copies an approved OA request into her recruitment tracker

**09:30**
Keys yesterday's joiner into SG HR, then into the Excel master

**11:00**
Works out which job title is right when SAP and SG HR disagree

**13:30**
Exports attendance and leave, and asks HQ for theirs on WeChat

**15:30**
Fixes HQ's garbled Chinese headers and YYYY/MM/DD dates

**18:30**
Leaves with the month's consolidation half done

> "We have four systems, and I am still the one connecting them, one spreadsheet at a time."

A proto-persona with an illustrative quote, built from the problem statement and desk research, to be validated with Organisation E.

ROOT CAUSE

# Digital, but isolated

Every system works on its own. Nothing connects them, so people do.




From Problem Statement 17: approvals happen in OA but the follow-up work happens elsewhere, and the Excel master is the only consolidated view.

WHAT SUCCESS LOOKS LIKE

# Enter it once, and let it flow

~16 h → **≤ 2 h**

HR hours a month consolidating attendance and leave

3–5 work days → **< 5 min**

From an OA approval to the onboarding checklist

~3 → **1**

Times each new hire's details are typed by hand

Weeks → **< 1 day**

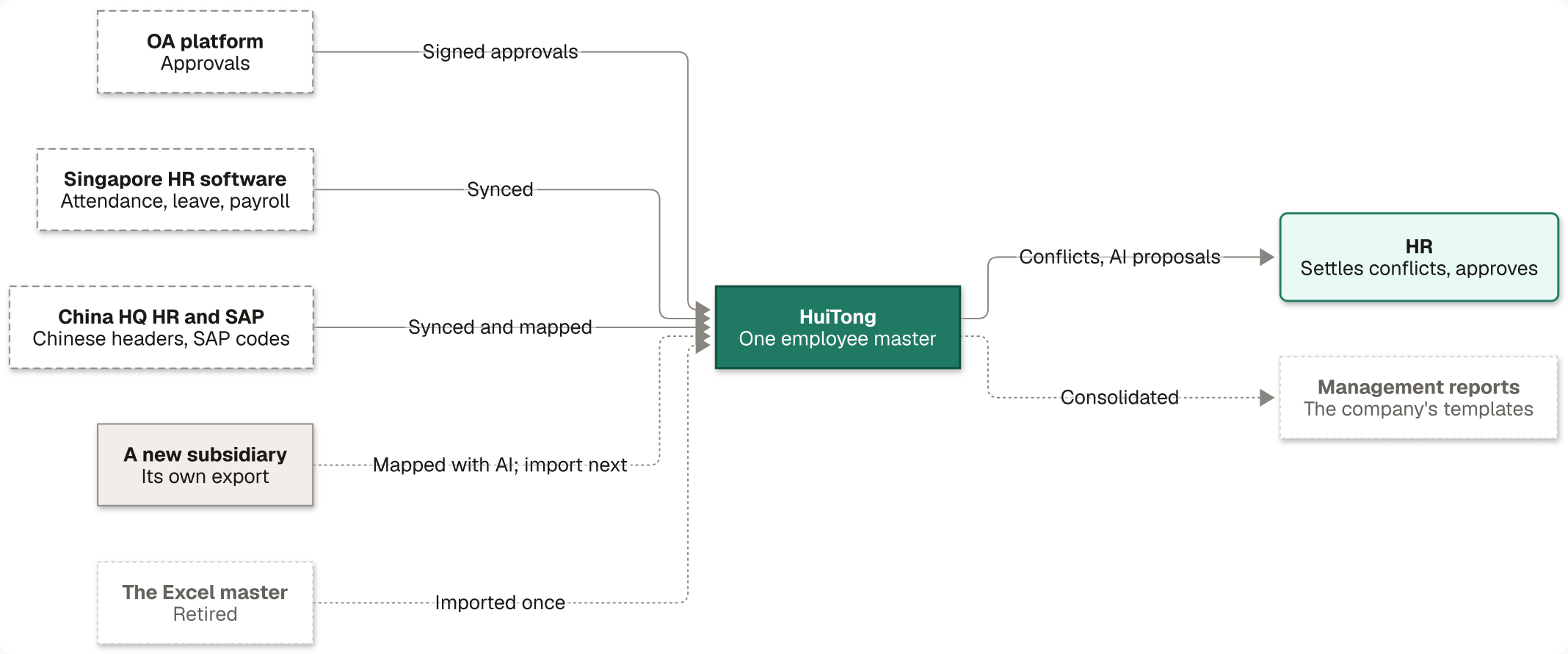
To connect a new subsidiary's HR export

The baselines are assumptions until we measure them with Organisation E. No gain may cost a guardrail: AI never writes without a person's approval, no national ID numbers are stored, and another entity's data is seen only as the transfer register allows.

THE SOLUTION

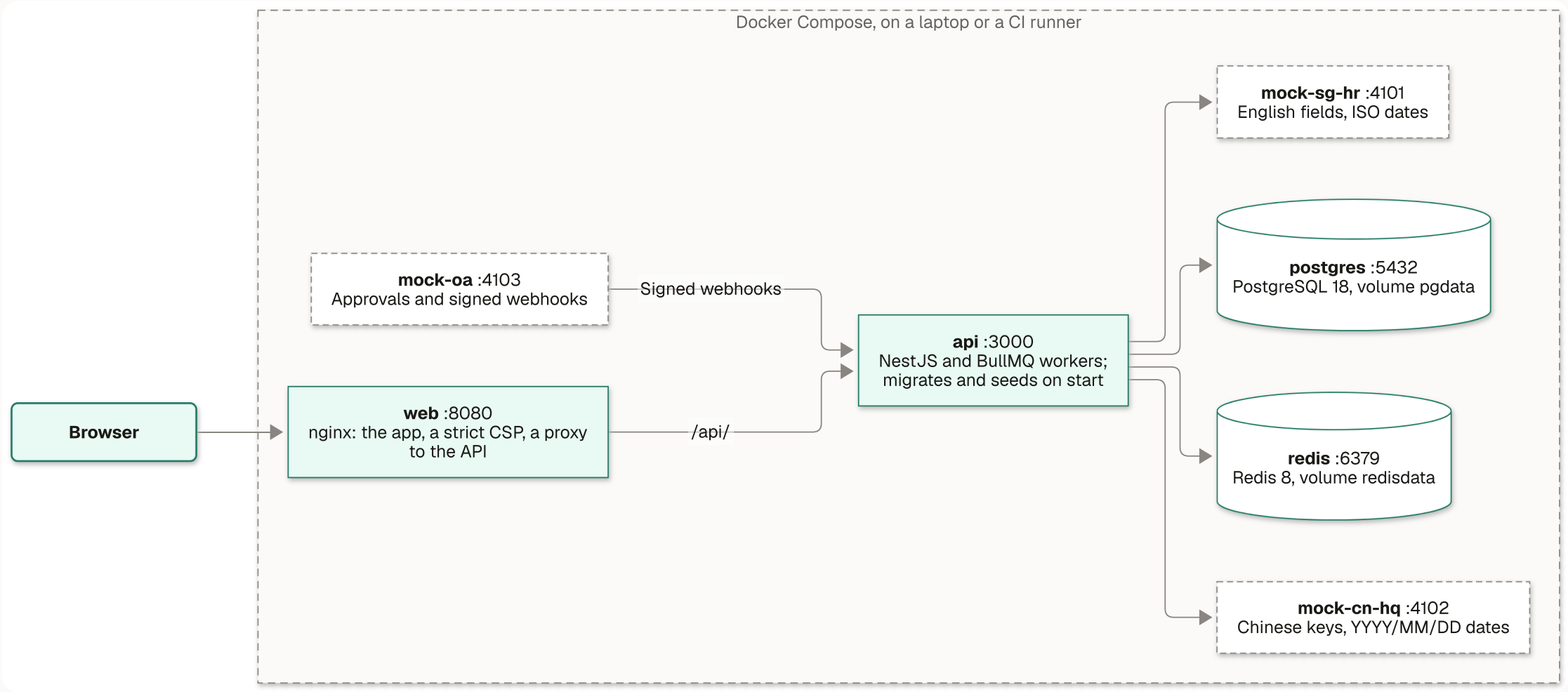
# An integration layer, not a replacement

Every system of record stays, behind one adapter each; only the Excel master retires



ARCHITECTURE

# One web app, one API, the systems around it



A KEY WORKFLOW

# One approval, four areas

Arjun's manpower request for a relationship manager, end to end

1

OA's signed approval arrives, and is stored once

Areas 1 and 2 · Receiving built

2

The requisition, the onboarding checklist and the preparation tasks appear

Area 2

3

A draft employee record, linked to the request

Areas 1 and 2

4

With the milestones met, SG HR's verified record activates the draft

Area 1 · Sync built

5

Reconciled with HQ's record by the group employee number

Area 1 · Matching built

6

Later leave imported and mapped: 年假, 病假, 调休

Area 3

7

HQ's report holds only the fields HQ may hold

Areas 3 and 4 · Area 1's allowlist built

8

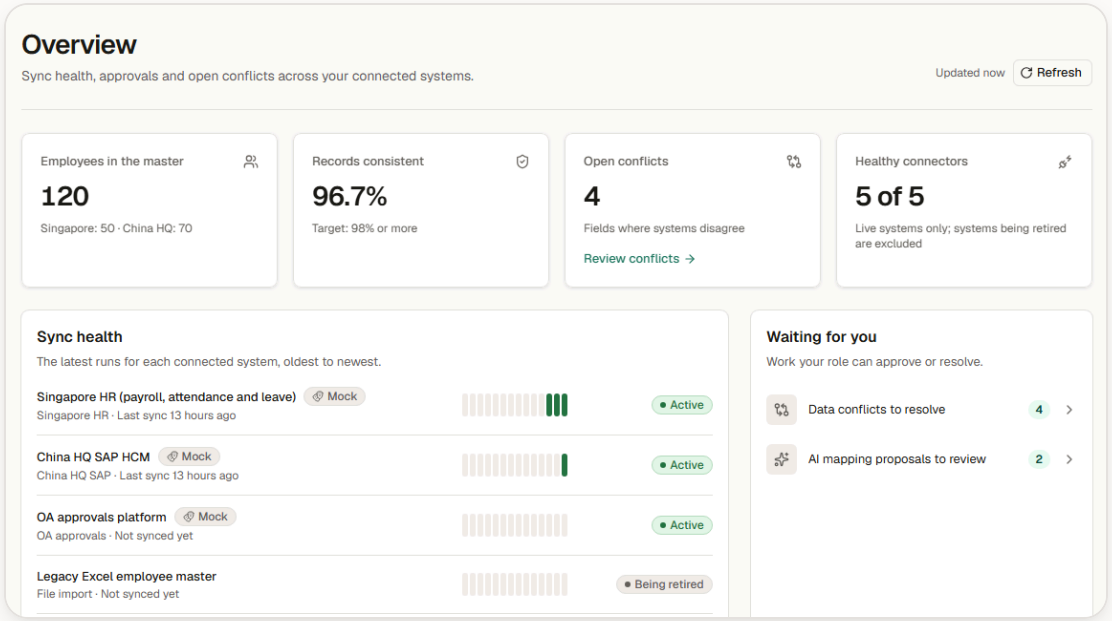
The overview shows the history and any exceptions

All areas · Overview built

Playwright runs this journey as E2E #1 on every pull request: step 1 runs today, and each other step is marked with its story until that story lands.

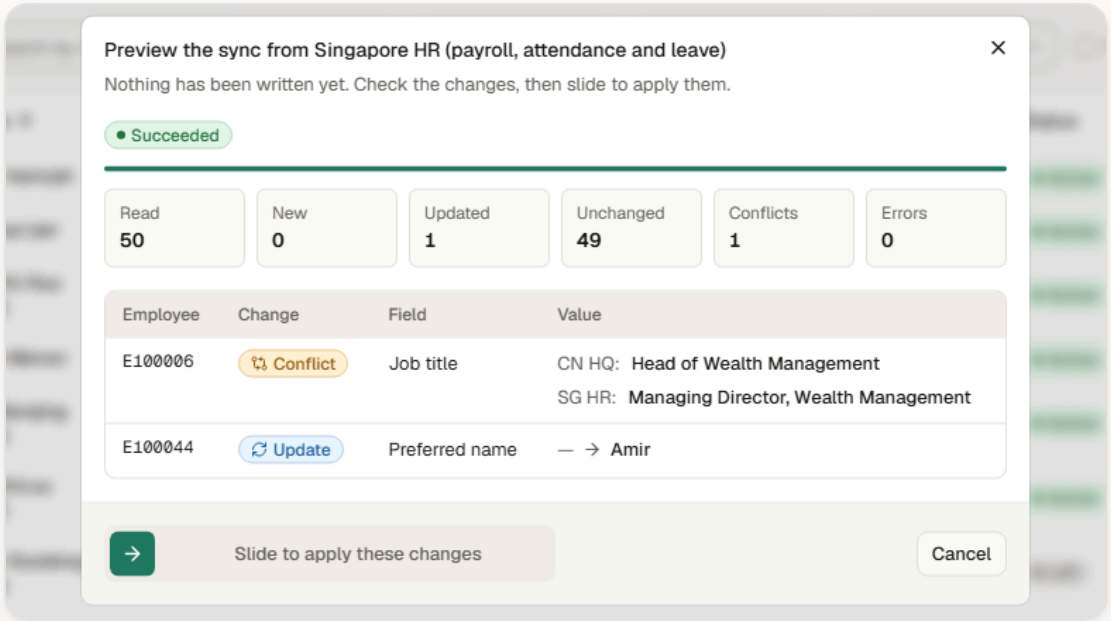
THE PRODUCT TODAY

# Built on the real API, with synthetic data




**The overview**

Sync health and the work waiting for HR; every mock is labelled



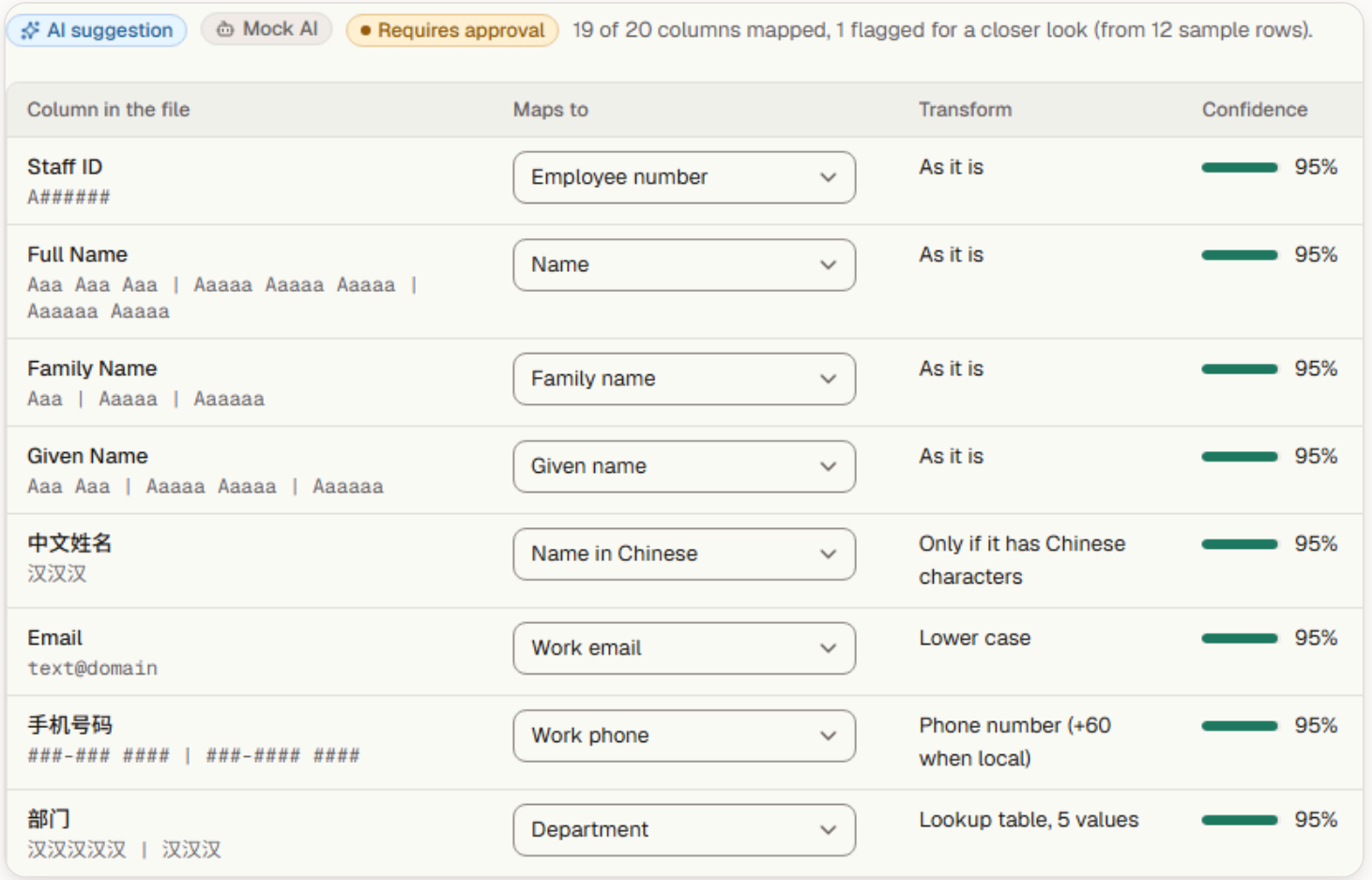

**A sync, previewed**

Nothing is written until HR has seen the changes and the conflicts

AI, WITH A PERSON IN CHARGE

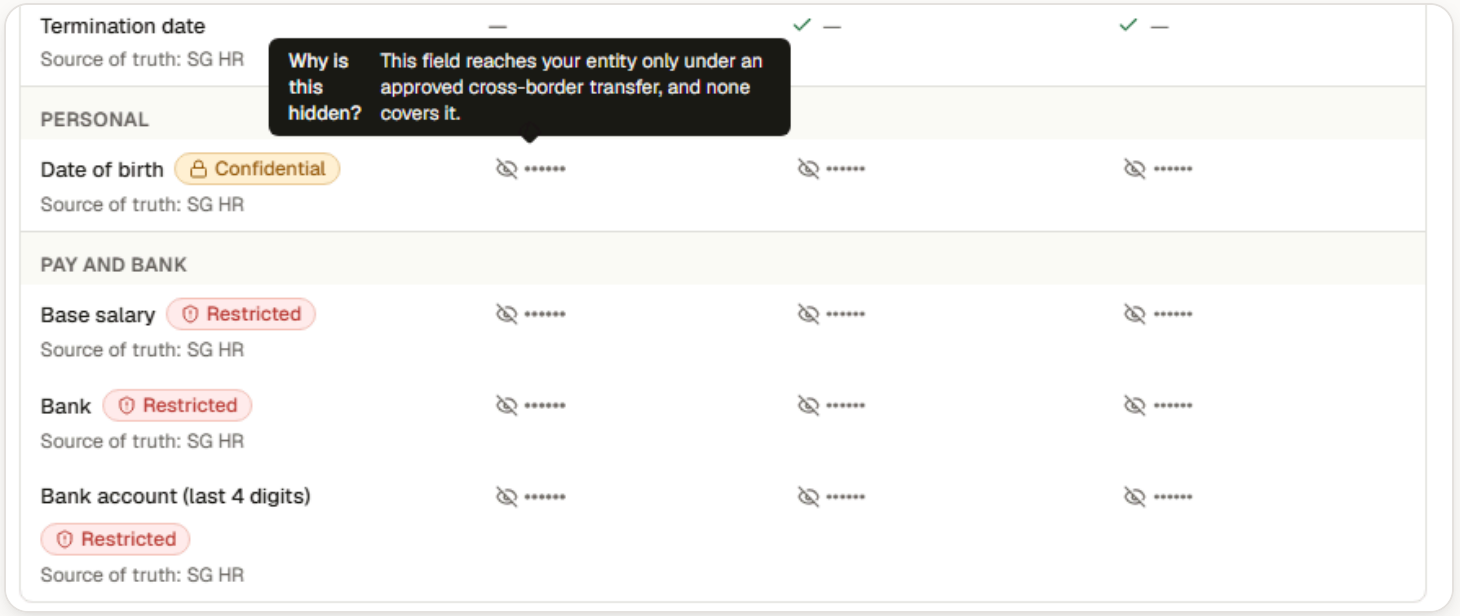
# The AI proposes; HR decides

- **Maps** columns to the master, Chinese headers too
- **Explains** disagreements, in English or 中文
- **HR decides,** and every approval is audited
- **No identifiers** ever reach the model
- **Every call** recorded: tokens, cost, outcome



# Controls built in, not bolted on

- **Least privilege:** seven roles, by entity
- **Masked fields** say why they are hidden
- **Preview first:** HR sees a sync before it writes
- **Audited:** every change to records, mappings and access
- **Cross-border:** default deny, by register




Seen by HQ HR: the date of birth withheld, with its reason, and the pay fields masked

DEPLOY AND MONITOR

# Watching it run: health, usage and AI cost

**Observability**
Requests, errors and the error rate

| Requests | Errors | Error rate |
|---|---|---|
| 94.7K | 296 | 0.31% |

**Clickstream**
First-party, from consenting sessions only

| Sessions | Page views | Views per session |
|---|---|---|
| 442 | 2,910 | 6.6 |

**AI usage**
Calls, tokens, estimated cost and latency

| Calls | Tokens | Estimated cost | Error rate | Median latency |
|---|---|---|---|---|
| 533 | 760.3K<br>585.3K in, 175K out | $4.38 | 3.2%<br>Failed calls and output that failed validation. A proposal HR rejected is not an error. | 50ms |

The dashboards show labelled sample data until area 4 builds the aggregations, in weeks 14 to 16.

NEXT

# Roadmap, and what we need to ask

**W5**
The proposal, on this prototype

**W8 to W12**
Each area builds its core logic

**W13**
The technical review

**W14 to W16**
Observability, clickstream and AI token tracking

**W18 and W19**
The final presentation, then the client check-in

## Questions for Organisation E

- Which HR products run in Singapore and at HQ, and what APIs or exports do they offer?
- Which OA product runs the approvals?
- Is there one group employee number in both systems?
- Where may employee data be stored and processed?
- May HuiTong write back to the systems of record?

THE TEAM

# Four areas, each owned end to end

AREA 1

## Integration core and platform

Connectors, the employee master, mappings and the AI assistant, syncs and conflicts, access enforcement, the platform and the integration map

Richie, team lead

AREA 2

## Workflow automation

OA approvals through to onboarding: the state machine, retries, SLAs and notifications

Owner to be named by W3

AREA 3

## Attendance, leave and reporting

Both HR systems' attendance and leave, the category mapping, the company template and reports

Owner to be named by W3

AREA 4

## Governance, security and monitoring

Administering access and the transfer register, the audit-log viewer, observability, clickstream and AI cost, and the AWS deployment

Owner to be named by W3

Richie, Iok Teck, Azri and Angelyn. From week 8 each member builds the area they own, and explains that code at W13 and W18.